

SkillClaw

Collective Skill Evolution for Multi-User LLM Agent Ecosystems

Let skills evolve collectively with agentic evolver

arxiv.org/abs/2604.08377

Agent Skills Remain Static After Deployment

Users Rediscover Solutions Independently

Current: Static Skills

- Skills manually installed from centralized hub
- Solutions discovered during interaction rarely persist
- Same failures and recoveries observed repeatedly
- Each user independently rediscovers solutions
- No knowledge accumulation at system level

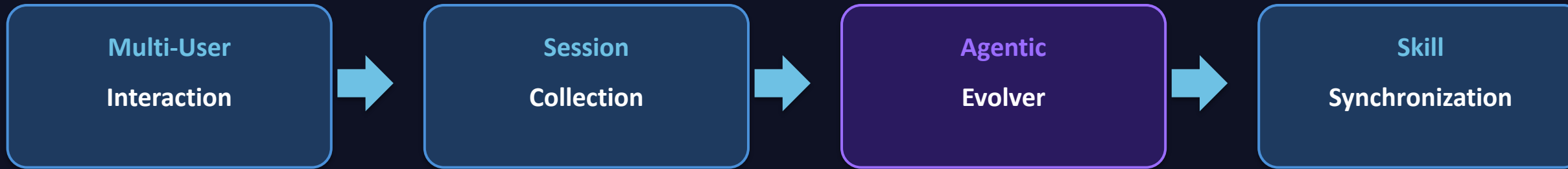


SkillClaw: Evolving Skills

- Runtime interaction drives skill improvement
- Cross-user trajectories aggregated as evidence
- Agentic evolver proposes validated mutations
- Shared ecosystem improves continuously
- Collective knowledge replaces isolated rediscovery

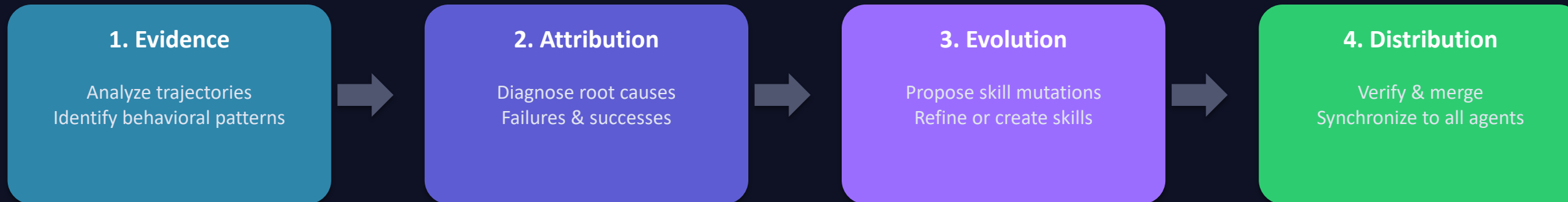
Memory-based methods remain instance-specific; skill-based methods treat libraries as static — SkillClaw bridges both gaps.

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution



← Closed Loop: Updated skills distributed back to all agents →

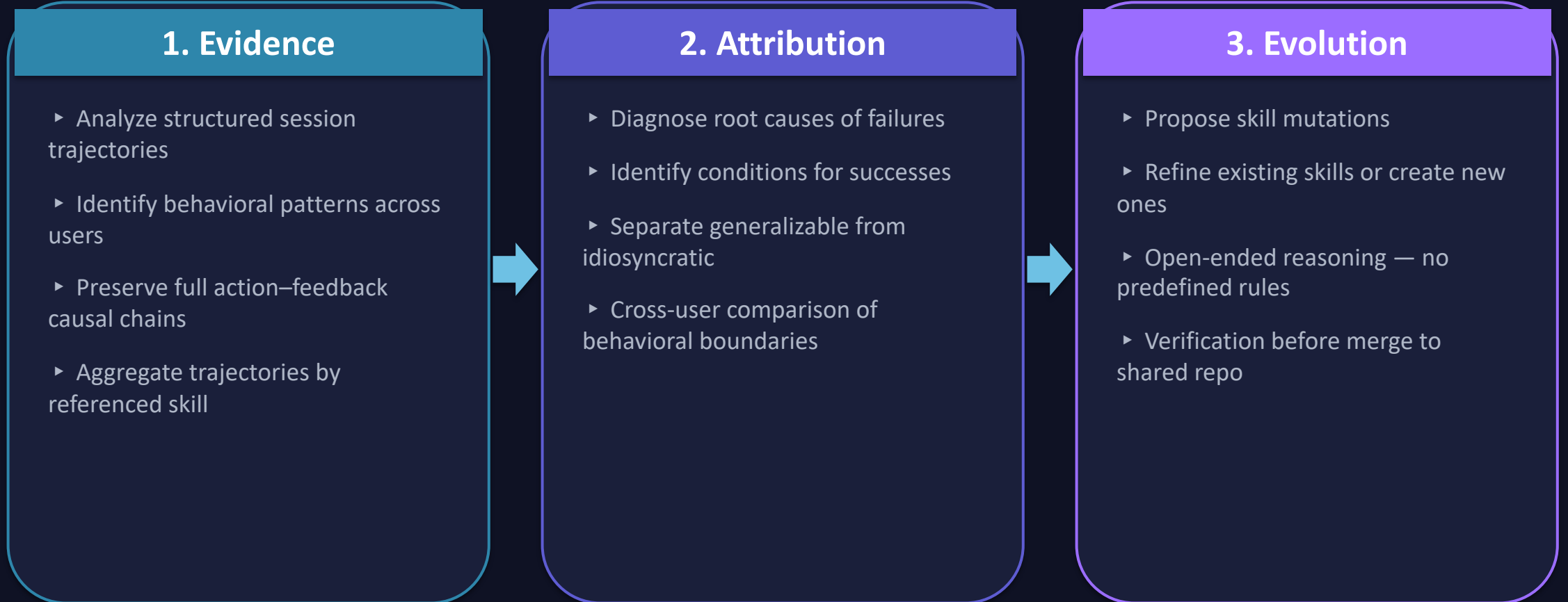
Agentic Evolver — Four Internal Stages:



The evolution process is driven by an autonomous agent performing open-ended reasoning over interaction evidence — not predefined update rules.

Agentic Evolver: Evidence → Attribution → Evolution

Three-Stage Pipeline Inside the Evolver



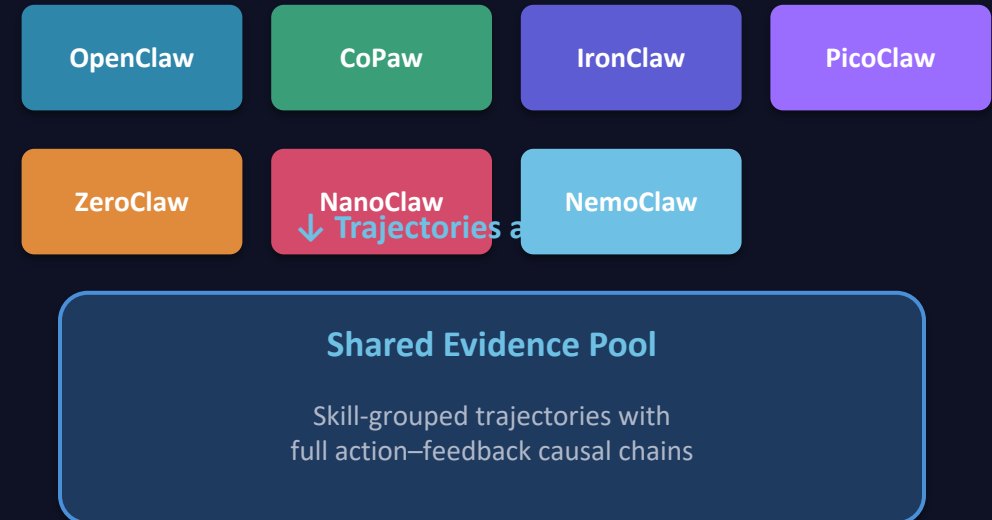
Key distinction: Skill updates are produced through open-ended agent reasoning over evidence, not predefined update rules or static heuristics.

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

Complementary Signals from Diverse User Contexts

- ▶ Different users exercising the same skill under diverse contexts produce complementary views of that skill's behavioral boundary
- ▶ Cross-user trajectories reveal both conditions under which a skill works and those under which it breaks
- ▶ A single user rarely generates enough signal to separate a generalizable improvement from an idiosyncratic fix
- ▶ Aggregation across users forms a shared evidence base for the evolver

Compatible Claw-Style Systems



Why Aggregation Matters

Individual user trajectories → idiosyncratic fixes that may not generalize

Cross-user trajectory aggregation → complementary behavioral signals that expose generalizable improvement opportunities

Evaluated on WildClawBench using Qwen3-Max as backbone — demonstrates substantial improvements across tasks.

Case Study: Multimodal Creative Pipeline Skill

6 Days of Evolution Attempts — 1 of 6 Accepted

Day	Candidate Skill	Skill Function	Validator	Next-Day Action
1	validate-tmp-workspace-inputs	Check /tmp_workspace inputs & environment setup	Accept	Upgrade to new best pool
2	multimodal-input-validation-and-setup	Multimodal input validation & output env init	Reject	Keep current best pool
3	multimodal-creative-task-pipeline	Multimodal creative pipeline (extract from PDF/video/image)	Reject	Keep current best pool
4	multimodal-creative-task-pipeline (impr.)	Added image classification, visual generation, structured validation	Reject	Keep current best pool
5	multimodal-creative-task-pipeline (impr.) + validate-required-input-files	Creative pipeline + per-file fail-fast validation	Reject	Keep current best pool
6	multimodal-creative-task-pipeline (cand.)	Extended PDF-to-poster / document-to-visual paths	Reject	Not admitted to next cycle

Conservative verification prevents regressions: only 1 of 6 candidate skills was accepted over 6 days. The validator rejected 5 candidates that would have introduced unvalidated changes to the shared skill pool.

Case Study: Git Push Auth-Fallback Skill

Iterative Refinement Over 6 Days — 3 of 6 Accepted

Day	Candidate Skill	Change Summary	Validator	Next-Day Action
1	git-push-with-auth-fallback	Safe fallback instead of blocking on push failure	Accept	Add to Safety best pool
2	git-push-with-auth-fallback	Unified patch/bundle filenames, reduced inconsistency	Accept	Keep updated best pool
3	git-push-with-auth-fallback + git-clone-to-directory	Auth-alternative paths & secrets audit; fixed subshell pitfalls	Accept	Keep current best pool
4	(none)	Same-pool retest; validator confirmed current pool as best	Accept	Continue same best pool
5	git-push-with-auth-fallback	Added "push hang treated as auth failure"; no improvement	Reject	Keep current best pool
6	git-push-with-auth-fallback	Added identity config & filename consistency; no improvement	Reject	Not admitted to next cycle

Iterative refinement works when evidence supports it: the git auth-fallback skill was successfully improved 3 times before reaching a plateau on Days 5–6, contrasting with the creative pipeline's conservative outcome.

Key Takeaways: Three Properties That Distinguish SkillClaw

From Existing Systems

1

Collective Evolution

Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem — not isolated per-user fixes.

2

Fully Automatic

Skill evolution is driven by runtime interaction without manual curation or explicit user intervention. No human in the loop required.

3

Agentic Evolution Paradigm

Skill updates are produced through open-ended reasoning over evidence, not predefined update rules — enabling context-aware mutations.

Implications for Continuously Improving Agent Systems

- Cross-user trajectory aggregation provides complementary behavioral signals that no single user can generate alone
- Conservative verification prevents regressions while allowing genuine improvements to propagate
- The agentic evolver's evidence-driven approach scales to multi-system ecosystems (OpenClaw, CoPaw, IronClaw, PicoClaw, ZeroClaw, NanoClaw, NemoClaw)